

Implementation and Evaluation of a Hybrid Classical-Quantum Blockchain Using GHZ-Based θ -Protocol Validation

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Abstract

Blockchain systems require a reliable consensus mechanism to establish agreement among distributed participants before transactions are committed to a shared ledger. This work presents a computational implementation of a quantum blockchain validation protocol based on entangled quantum states, integrating the protocol with a classical blockchain transaction and ledger framework. The implementation models six validator nodes as participants in a six-qubit Greenberger–Horne–Zeilinger (GHZ) state. For each candidate transaction, one validator is randomly selected as the verifier, while measurement angles are generated according to the protocol constraint that their sum is an integer multiple of π . Each validator performs the corresponding quantum measurement and returns a binary measurement outcome. Instead of using conventional majority voting or independent approval from validators, the verifier determines consensus from the XOR parity of the collected measurement outcomes and compares it with the parity prescribed by the measurement-angle condition. The implemented system combines this quantum validation stage with classical blockchain operations, including SHA-256 transaction hashing, block generation, validator records, and replication of accepted blocks across a shared ledger. The transaction hash is retained for classical data integrity, while quantum measurement outcomes are used specifically for the consensus decision. The quantum execution layer is separated from the blockchain application through a backend interface, enabling the protocol to be executed using

a Qiskit-based quantum simulator. The resulting system provides a practical proof of concept for reproducing and integrating the quantum validation procedure described in prior work within an executable blockchain environment. It also establishes a software framework that can be extended for experimental evaluation and eventual execution on quantum hardware. Across 20 independent simulation trials, the quantum validation achieved a 100% parity success rate under ideal simulation conditions, with an average execution time of 3.25 ± 0.05 s per trial.

Keywords

Quantum Blockchain, GHZ state, Quantum Consensus, Blockchain Validation, Quantum Computing

1 Introduction

Blockchain technology ensures distributed state preservation, in which each of several players contributes to maintaining a common ledger without relying on a single governing authority. In a classical blockchain, there is a combination of different components that act together to achieve coordination. These components include transaction representation, cryptographic hashing, authentication, communication among the participants' nodes, transaction confirmation, achieving consensus, transitioning the state, and maintaining the distributed ledger. When a correct transaction is confirmed, it is put into the block according to the consensus and validity rules, and then the state of the blockchain is updated and preserved by the participants. Thus, the blockchain's reliability is a product of the interactions among different elements of the distributed system rather than the effectiveness of a particular cryptography method.

The principles of blockchain originate from previously existing ideas in area of cryptography and decentralized digital systems. In 1991, researchers Haber and Stornetta presented a solution for cryptographically linking timestamps of digital documents which became the paramount idea of tamper proof series of records [1]. Then there were a number of ventures such as b-money and Bit Gold which were aimed at exploring decentralized digital values and means of recording digital processes without needing the existence of a central authority. The concepts elaborated in the course of these developments later proved to become the building blocks of blockchain techniques and included cryptographic integrity, decentralized activities, distributing records and shared states.

A key change emerged with the release of Satoshi Nakamoto's Bitcoin: A Peer-to-Peer Electronic Cash System [2] and the start of the Bitcoin system in 2009. The introduction of Bitcoin included features like the linking of a hash, digital signatures, peer-to-peer technology, transaction validation, and proof of work making Bitcoin a working decentralized ledger system. Hence, the importance of this development exceeds the limits of the chained blocks allowing us to see how transaction integrity,

communication, consensus, and replicated state can work without a central administrator. After that, blockchain technology developed into something more than just money transfer systems. Ethereum developed in 2013-2014 and was released in 2015, expanded that system by adding a function of smart contracts making blockchain not only a method of transferring money but rather a platform for decentralized applications.

The classical blockchain framework creates the basis upon which quantum blockchain studies have evolved. Specifically, aspects like transaction validity, participation of validators, consensus, state transformation, and maintaining the ledger still apply, even when quantum technology is involved. Nonetheless, quantum computing has added a new feature to the problem of blockchain studies, since quantum computing changes the assumptions regarding the security of classic blockchains and also allows using quantum data as a resource in blockchain systems.

Quantum computing relies on the manipulation of quantum states which have characteristics like superposition, interference, and entanglement. The creation of quantum algorithms has shown that quantum computers could offer computational benefits for certain types of problems. The algorithm proposed by Shor in 1994 proved that integer factorization and discrete logarithms could in principle be solved efficiently on a quantum computer that is sufficiently capable [3]. As these problems form the basis of important public-key cryptographic systems, the finding indicated a possible long-term danger to digital signatures and to other cryptographic methods used in blockchain systems. Grover's algorithm, which was introduced in 1996, showed a quadratic speedup for unstructured search, the complexity being approximately $O(\sqrt{N})$ as opposed to $O(N)$ for a simple classical search [4]. While the implications of Grover's algorithm are different from those of Shor's, it is still relevant to blockchain since hashing and computational search are part of several blockchain security and consensus mechanisms. As a consequence, the tie between quantum computing and blockchain took form along two main lines described above. One direction is defensive and aims at ensuring the safety of existing classical blockchain solutions from future quantum attacks. The other direction is constructive and studies the possibilities of working quantum computation, telecommunication, and quantum mechanics themselves in the realm of blockchain technology. The first direction led to studies of post-quantum blockchain systems and quantum-protected blockchain architecture, while the second direction gave rise to quantum-augmented and purely quantum blockchain technologies. Kiktenko et al. presented one of the first responses to quantum threat by maintaining the classical blockchain structure but changing some components to increase their security. Kiktenko et al. considered quantum-protected blockchain architecture containing Quantum Key Distribution together with an ordinary blockchain setup. The findings showed the feasibility of using Quantum Key Distribution via urban fiber networks, implying that quantum communication can ensure security without the need to turn transactions, blocks, or records into quantum states [5].

Another different course is post-quantum cryptography (PQC) wherein cryptographic primitives that are exposed to quantum attacks have been substituted by those that can withstand the quantum attacks that gave birth to the QuantumShield BC which incorporates quantum random number generation technology, quantum key distribution technology and quantum Byzantine Fault Tolerance [6] as well as utilizing Dual Layer Signature Mechanism [7]. Hence, both of their approaches aim to tackle the quantum threat while preserving a basic classical blockchain system. It is essential to note that there is a difference: quantum-secured blockchain uses quantum technology applied to one or more of the relevant security functions whereas post-quantum blockchain changes the cryptographic premises of classical technology regardless of quantum technologies employed.

This development brings us to an important question: instead of using a quantum technology purely for the protection of a classical blockchain, can quantum information itself come into use for representation, validation, or becoming a part of the consensus mechanism of the blockchain? This question represents the emergence of fundamentally quantum or quantum-native blockchain architectures, where quantum state, quantum correlations, phase information, or entanglement will be used to represent information, validate blocks, or decide whether a transition should be accepted.

As a result, there has been considerable growth in various threads of research based on quantum blockchain technology. One such research area deals with the concept of quantum block design, with an emphasis on mapping classical blockchain data onto quantum states. A recent study of Banerjee, Mukherjee, and Panigrahi focused on quantum blockchain technology that is based on quantum hypergraphs and which incorporates phase-based methods for quantum representation of blockchain information. Their research has raised a number of significant questions on the questions of transforming classical data into quantum states and establishing relationships between the subsequent states [8].

A different aspect is instead oriented to achieving quantum consensus and validation based on entanglement. In ordinary blockchains, the nodes that participate in the blockchain can apply their own validation to a transaction and thus validate it through the vote or any form of agreement. The entanglement method is about the question of whether from the correlations existing among the measurements done by quantum systems, evidence can be obtained confirming that the participants have a legitimate quantum state. In this respect, GHZ states are very important because they provide for the correlations involving different quantum systems. Each qubit can be associated with a participating node, and their results can be studied to see whether the measured correlations correlate with the expected one.

Consequently, this indicates a significant shift in the validator's function. In a classical validation process, Transaction→Individual Validation→Vote/Approval, while an entanglement based procedure can be shown as Shared Quantum State→Distributed Measurement→Correlated Outcomes→Quantum Validation Condition. Thus, the

validator can be thought of as not being an independent voter. The validator becomes a participant in a quantum measurement process with the final decision based on the quality of information between measurement outcomes, which is one of the key points of this work. In addition to these quantum-native method, hybrid quantum-classical architecture is being actively developed since the current state of quantum hardware and quantum networking cannot perform all required functions of a classical distributed ledger. In hybrid architecture, classical systems can do everything from managing transaction data, hashing, metadata, networking, application logic, and persistent ledger storage, while quantum resources involved for separate functions like state preparation, entanglement generation, validation, authentication, and signatures. This means that “quantum blockchain” is not a term for a well-defined architecture but may refer to various replacements of the conventional blockchain architecture or its components with quantum technology. In other words, it can be introduced in order to ensure protection against quantum attacks, transmission of quantum messages, creation of quantum data, verification of the states of the system including that it has achieved consensus, etc. Thus, putting a quantum technology into a blockchain system is not a sufficient condition for saying that this system is quantum.

During this development, the work of Del Rajan and Matt Visser, titled Quantum Blockchain Using Entanglement in Time,[9] serves as the basis for this study. The work was first uploaded to arXiv in 2018 and then published in Quantum Reports in 2019. The main issue is that quantum entanglement is applied as one of the building blocks of blockchain but not just as a means of increasing security. The suggested construction is based on the concept of a temporal Greenberger–Horne–Zeilinger (GHZ) state that builds correlations among quantum systems that are related in time and incorporates those correlations into the blockchain structure.

The Rajan-Visser framework is important because it establishes a link between quantum entanglement and the process of blockchain validation. In accordance with the framework, an individual among the nodes engaged in the process is assigned the role of a verifier, and the quantum block is distributed to all the participating nodes. The verifier presents measurement parameters, following the premise of the protocol, and each of the other participating nodes executes the measurement on its own quantum instance. Afterwards, the nodes send a binary result of their measurements to the verifier, and the verifier processes their results according to the XOR/parity condition based on the measurement angles.

The importance of this mechanism rests on three interconnected components: Shared Quantum State→Distributed Measurements→Collective Validation Condition. The shared quantum state creates correlations among the participating network nodes, the distributed measurements provide a method of turning quantum signals into classical information, and the parity condition translates the correlated classical information into a blockchain-level validation.

Simultaneously, the work done by Rajan–Visser provides a conceptual framework for validation rather than an actual application of a traditional blockchain network. The authors indicate the need of development of low-level consensus and quantum network implementation details. This distinction creates a crucial implementation-level background for further work: the conceptual mechanism explains how a quantum correlation can be associated with validation while the engineering problem is how to connect this mechanism to transaction processing, validator’s management, block generation as well as ledger’s status and the real quantum execution.

The current research investigates this implementation and integration layer. Instead of introducing a new entanglement-based consensus method, this research creates an executing hybrid environment of blockchain that is based on the quantum verification process as described in the Rajan-Visser framework. This serves as a foundation for creating an executable workflow of transition from quantum and classic verification processes. The contribution made in the present study can therefore be regarded as an implementation study of a classical quantum verification technique. The proof-of-concept thus created forms grounds for future examinations with regard to quantum noise, multiple measurements, dishonest or defective participants, different validator configurations, scalability, and depositing on a real quantum machine.

2 Methods

A hybrid classical-quantum blockchain prototype was built here. Conventional blockchain tasks like transaction handling, hashing, block assembly, and ledger copying stay unchanged, but validation uses an entanglement-based quantum method. This relies on the Theta-protocol from quantum blockchain studies, especially the GHZ-state check by Rajan and Visser. Six validators act as six people sharing a six-qubit GHZ state. For every new transaction, one validator Picks randomly to verify it, and six measurement angles follow the needed angle rule. One shared GHZ circuit forms, measurements hit all six qubits, giving one bit result per validator. These six bits go back to the blockchain part, where the checker tests the protocol’s XOR parity rule. Only if that rule passes does the transaction move forward to make a block and lock into the shared ledger.

We retain the classical blockchain mechanisms wherever necessary. For example, we use classical mechanisms such as SHA-256 processing on each transaction to generate a deterministic “transaction hash” that supports classical data integrity/transaction identification mechanisms. Our use of the quantum mechanism is limited to the validation (or consensus) stage; we do not replace classical hashing with our quantum approach. In particular, the numeric representations derived from the SHA-256 transaction hash differ significantly from the angle measurements generated via the qubit’s behavior under the Θ -protocol. In other words, we treat the transaction hash as part of the classical blockchain, while ensuring quantum measurement angles can be generated independent of the transaction hash value. These quantum measurement angles

are created based upon the requirements for successful operation of the quantum validation process itself. Thus, we preserve the ability to separately evaluate performance aspects of the classical and quantum components.

At a high level, the classical layer handles aspects such as transaction processing, hashing, block building, and ledger management. Quantum components such as GHZ-state preparation, distributed measurements, correlated outcomes, and their corresponding validation condition are handled by the second layer of this structure. Finally, the actual execution of the quantum component itself — including all the elements of the computation discussed above — is decoupled from the rest of the blockchain application via a dedicated backend interface. This allows one (for example) to use a Qiskit-based quantum simulator that will run the circuit on behalf of the system and provide results (i.e., the measurement outcomes) back to the blockchain layer over an API connection. This allows a modular architecture where subsequent blockchain-level workflows could potentially plug into other quantum backends without having to redo all of the ledger implementation. In this case, the implementation is a computational proof of concept instead of an actual realization of a quantum network. The paper on which the project is based explains the quantum blockchain as a conceptual design that uses GHZ-state verification as a method of block verification.

2.1 Experimental Environment

2.1.1 Computing Platform

The goal of the experiment is to assess the blockchain process and the quantum verification procedure without relying on cloud quantum technology. Therefore, the actual running of the circuit is done locally with the help of Qiskit Aer. In that way, the experimental environment consists of two layers cooperating with each other: Classical Application Layer + Local Quantum Execution Layer. In the classical layer, the transactions, validators, blocks, consensus state, and ledger synchronization are controlled, while the quantum layer develops and runs the six-qubit verification circuit and returns the six measurement data for block acceptance assessment.

2.1.2 Software environment

The principal software components used for the quantum implementation are given in Table 1.

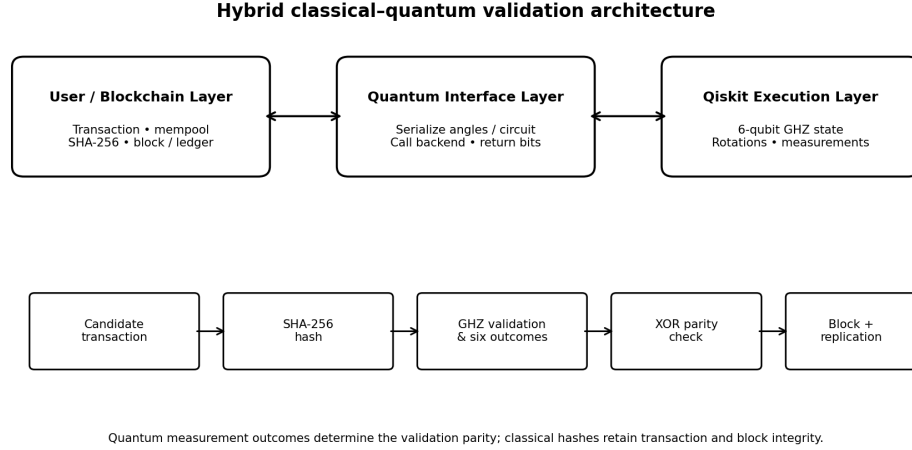
Quantum execution code specifically imports QuantumCircuit from Qiskit and AerSimulator from Qiskit Aer. The simulator backend accepts six measurement angles, constructs a six-qubit circuit, executes it in one shot, and returns six binary measurement results. Use of local Aer backend is intentional. No IBM Quantum hardware was utilized in the experiment and no cloud execution was needed for the consensus Pipeline. Thus, hardware queue time, cloud-access requirements, and hardware-related noise were removed from the baseline version.

Table 1: Principal components used in the proposed system

Component	Version / Configuration	Role
Python	Python environment used for the Qiskit backend	Quantum execution
Qiskit	2.2.3	Quantum circuit construction
Qiskit Aer	0.17.2	Local quantum simulation
NumPy	2.4.6	Numerical operations
SciPy	1.17.1	Scientific-computing environment
HTML5	–	User-interface layer
CSS	–	User-interface presentation
JavaScript	–	Blockchain application and control logic

2.2 System Architecture

The entire system architecture is comprised of three major layers which is described in the Figure 1. The figure also gives a expanded view of the overall workflow that happens within the 3 layers.

**Fig. 1:** 3 Layer Architecture

The user and blockchain layer consists of the transaction model, mempool, validator nodes, blocks, public ledger, conventional hash functions, and consensus mechanism. The quantum interface layer is responsible for transformation of the blockchain-based measurement parameters to the serialized six-qubit circuit description and invokes the quantum backend. The Qiskit execution layer forms the actual

QuantumCircuit, prepares the GHZ state, performs local angle-dependent measurements, runs the circuit using AerSimulator and obtains six measurement results. This separation is reflected in the architecture of the software.

2.3 Blockchain Network Configuration

As illustrated in Figure 2, our blockchain has one primary user node with six validator nodes (denoted by $V = V1, V2, V3, V4, V5, V6$). These six validators have their unique identifiers, network addresses, states and their own copies of the blockchain ledger. We map these six validator nodes to the six qubits in our quantum verification circuit where $V_i \leftrightarrow q_i$ for $i \in 1, \dots, 6$. Therefore, there are six validators $\equiv$ six quantum participants $\equiv$ six qubits. To emphasize this, note that all the validators are not individual quantum circuits but only part of one large shared quantum circuit – thus, having six quantum participants means they share one big quantum circuit representing the collective verification state. Also, each quantum participant (validator) can give us one outcome of the measurements made on them (Y_i).

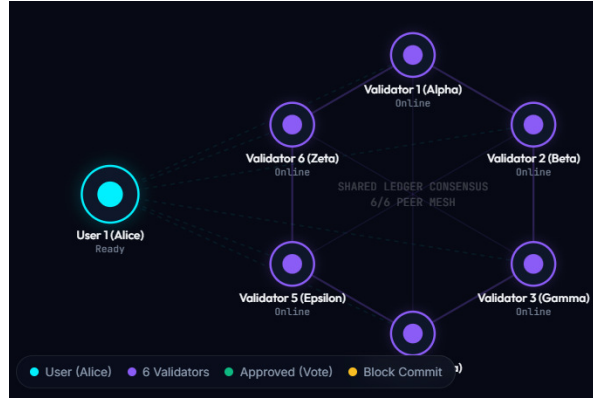


Fig. 2: Blockchain Network Arrangement

Our block chain has various different state structures including Shared Ledger, Mempool and Local Validator Ledgers. The Shared Ledger shows up as an authoritative source of truth held within the application itself. While every validator holds their own synchronized copy of the blockchain's ledger. In order to avoid concurrent processes of more than one transaction we implemented a lock around the process which allows us to process just the very first transaction in the Mempool before moving on to the second.

2.4 Genesis Block Initialization

Prior to starting normal transaction processing, the blockchain needs to be initialized via a genesis block. A genesis transaction gets generated internally, which sets an initial balance to the primary user. Its SHA-256 transaction hash gets computed. Its

status gets marked as committed, and based on that, its corresponding genesis block gets created having zero-value for its previous-hash field. This genesis block will get inserted in the shared ledger and simultaneously replicated across the local ledgers of all validators. Thus, this genesis block is primarily utilized to initialize/establish the initial common states of all six- validator networks. The protocol is such that $B_0 \rightarrow \{L_1, L_2, L_3, L_4, L_5, L_6\}$. (In other words, the quantum Θ -protocol isn't used for creating this initial bootstrapPing state - the quantum validation mechanism will be applied to subsequently submitted candidates.) By having this property, we prevent our initialization phase from requiring an invalid consensus state. To ensure this happens correctly, our implementation explicitly synchronized the genesis block with all 6 validators prior to allowing normal transaction submission.

2.5 Transaction flow

Transaction initiation begins from the perspective of the primary user who submits the transaction containing relevant parameters such as sender, recipient, amount to send, transaction fee, timestamp, metadata, and transaction identifier. Prior to insertion into our validation Pipeline, basic classical checks are run ensuring positivity of the amount parameter and that the sender has sufficient balance to pay for both the amount and transaction fees (in order). Once validated, this transaction will be placed into the mempool which serves as a temporary storage system for transactions which have been created but not yet committed to the blockchain's stateful ledger. For simplicity reasons during our experiments, we've designed the Pipeline to only handle processing of one transaction at a time as shown in Eq 1,

$$T_0 \rightarrow \text{Mempool} \xrightarrow[\text{Classical}]{\text{Quantum}} \text{Block } T_0 \quad (1)$$

Such an approach ensures that we may cleanly associate all results directly attributable to the quantum validation with just the one candidate transaction. Prior to running quantum validation on a transaction, it was required to compute its hash in accordance with SHA-256 producing a 256-bit digest representing the contents of the transaction. In addition to the hexadecimal encoding provided natively by the algorithm, we've recorded the same result using the binary and decimal bases. To avoid losing any bit due to JavaScript rounding issues when storing large integers, the latter is computed via BigInt instead of relying upon the native Number class. It should also be noted that when considering the binary format of the hash, care is taken to maintain the exact 256 bits of output including leading zeros. Finally, we derive a classical transaction associated angle defined as:

$$\Theta_{\text{tx}} = \frac{D}{2^{256} - 1} * \frac{\pi}{4}, 0 \leq \Theta_{\text{tx}} \leq \frac{\pi}{4} \quad (2)$$

From Eq 2, D represents the decimal equivalent value of the given 256-bit transaction hash. This guarantees the resultant angle falls within $[0, \pi/4]$. It should be noted that the value above is never considered part of the set of quantum measurement angles Θ_j participating in the consensus protocol. This distinction ensures that the quantum protocol remains separate from the classical tx_hash representation. Consequently, the transaction-associated parameter differs from the measurement angles, i.e., $\Theta_{\text{tx}} \neq \Theta_j$.

Once the transaction hash has been generated, one of the six validators is randomly selected to act as the verifier for the current validation round. Let the randomly selected validator index be defined as:

$$k = \lfloor 6r \rfloor, \quad r \sim \mathcal{U}(0, 1) \quad (3)$$

Thus, in Eq 3, $k \in \{0, 1, \dots, 5\}$, and validator V_k acts as the verifier for the current transaction. It should also be noted that the remaining validators are not merely independent classical voters; rather, each validator acts as a quantum node associated with its respective qubit of the shared GHZ state.

2.6 Generation of the Six Quantum Measurement Angles

In this case we need six measurement angles. Implementing this configuration you will generate 5 random positive values which are going to be used to define the measure angles, making sure these five angles cover an approximate 95% of the interval associated with π . For the last one we have calculated its value considering the rest of the range.

Let $r_1, r_2, \dots, r_5$ be five random values generated by the implementation, and let

$$R = \sum_{i=1}^5 r_i, \quad \alpha = \frac{0.95}{R} \quad (4)$$

In Eq 4 we have the sum of the 5 random values and a scaling coefficient factor is defined by α .

The first five measurement angles are calculated by Eq 5

$$\Theta_i = \Pi * \alpha * r_i, \quad i = 1, \dots, 5. \quad (5)$$

The final angle is calculated by the Eq 6

$$\Theta_6 = \Pi - \sum_{i=1}^5 \Theta_i, \quad \sum_{i=1}^6 \Theta_i = \Pi \quad (6)$$

We have implemented up to the normal floating point representation. To enforce the six angle constraint we have made this restriction explicit. I.e., the function will reject configurations containing anything but 6 validators. Given that the sum of the six measurement angles is constrained to equal Π , the right-hand side of the Θ -protocol parity condition becomes 1 as defined by Eq 7

$$\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \bmod 2 = \frac{\Pi}{\Pi} \bmod 2 = 1 \quad (7)$$

Therefore, we expect the six-validator experiment to produce an output satisfying the following property:

$$\left(\frac{1}{\Pi} \sum_{i=1}^6 \Theta_i \right) \bmod 2 = 1 \quad (8)$$

This rather strange-looking design of Eq 8 leads to an extremely simple final consensus condition of just requiring that the XOR of these six measured binary outcomes equals 1.

2.7 Construction of the Six-Qubit GHZ State

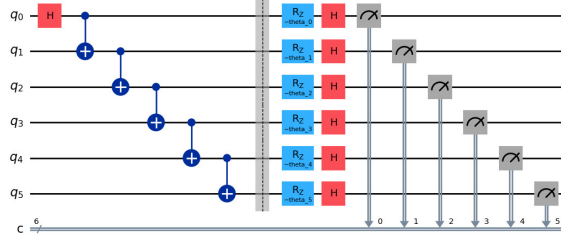


Fig. 3: Qiskit Implementation of 6 qubit GHZ circuit

As in the Figure 3, Quantum validation circuit has six qubits and six classical measurement bits ($q_0, q_1 \dots, q_5$) and ($C_0, c_1 \dots, c_5$). Initially these are set to $|000000\rangle$.

Then applying a Hadamard gate on the qubit q_0 we get Eq 9,

$$|000000\rangle \rightarrow \frac{1}{\sqrt{2}}(|000000\rangle + |100000\rangle) \quad (9)$$

Next applying a sequence of linear CNOT entanglement chain of gates yields the desired six qubit GHZ state which gives Eq 10,

$$|GHZ_6\rangle = \frac{1}{\sqrt{2}}(|000000\rangle + |111111\rangle) \quad (10)$$

Instead of implementing six independent circuits for each participant as would be expected from an ordinary GHZ circuit, it's implemented here as a single quantum circuit that creates a commonly shared quantum state between six validators.

2.8 Final Integration into the classical flow

The quantum-validation stage performs measurements in the Θ basis by applying an $R_z(-\Theta_j)$ operation followed by a Hadamard gate H to each validator qubit, yielding a binary measurement result $Y_j \in \{0, 1\}$.

Once computed, the browser application transmits the six measurement angles and the circuit description to the Python-based quantum backend over the network connection. After processing these inputs, the quantum backend returns the resulting six-bit vector as defined in Eq 11, whose result can be mapped to the corresponding validators.

$$\mathbf{Y} = [Y_1, Y_2, Y_3, Y_4, Y_5, Y_6], \quad V_i \leftrightarrow (q_i, \Theta_i, Y_i) \quad (11)$$

These measured bits represent a single quantum-validation event rather than independent approvals. Accordingly, consensus is calculated using the measured parity by Eq 12, where $\oplus$ denotes the exclusive OR (XOR) operation.

$$P_{\text{measured}} = Y_1 \oplus Y_2 \oplus Y_3 \oplus Y_4 \oplus Y_5 \oplus Y_6 \quad (12)$$

The measured parity is compared against the protocol condition and measurement angle constraint which gives $P_{\text{expected}} = 1$ by Eq 13.

$$P_{\text{expected}} = \left(\frac{1}{\Pi} \sum_{j=1}^6 \Theta_j \right) \bmod 2, \quad \sum_{j=1}^6 \Theta_j = \Pi \quad (13)$$

Therefore, for the transaction to transition from the **PENDING** state to the **VALIDATED** state, the measured parity must satisfy the Eq 14.

$$P_{\text{measured}} = P_{\text{expected}} = 1 \quad (14)$$

Finally, after storing the quantum-verification results along with the transaction data itself within our system, the next step involves generating a classical block consisting of the current transaction, its associated previous block hash, and transaction hash, as well as both the proposer and the records of verification provided by each validator. This last block can subsequently be secured using SHA-256 hashing. Importantly, though, one should note that the role of proposer comes from deterministic selection among those participating as validators; importantly, this position differs from that randomly chosen verifier who participated during the quantum validation/verification process. With this final step completed, the proposed block becomes part of the shared ledger and is distributed to all participants in the six-validator network. Ledger updates propagate across the distributed system by ensuring that each participant receives an identical copy of the validated block before appending it to their respective local ledger.

Following block propagation, user's balances are updated accordingly, and the contents of the transaction mempool are synchronized with the latest ledger state. To maintain system integrity and ensure correct operation, additional validation checks verify the relationships among the previous block hash, the current block hash, and the transaction hashes. These checks help ensure consistency across the distributed ledger and detect discrepancies between local ledger copies.

3 Results

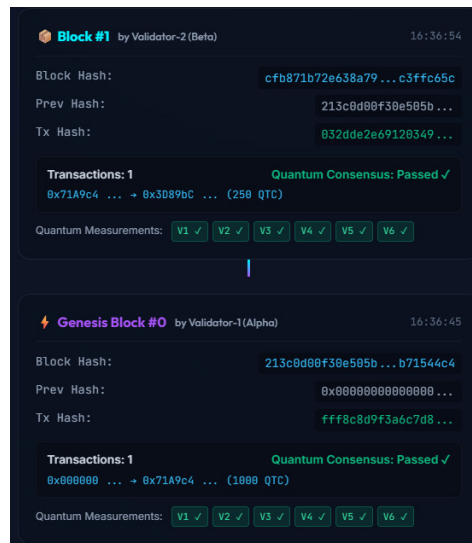


Fig. 4: Shared Ledger

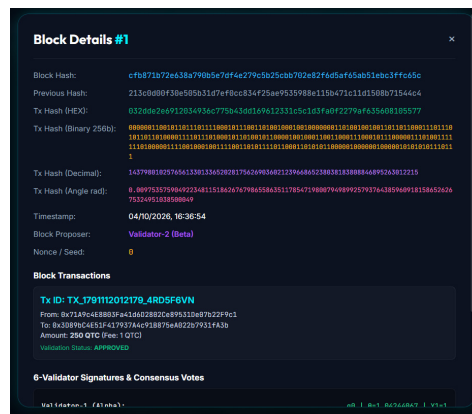


Fig. 5: Block information

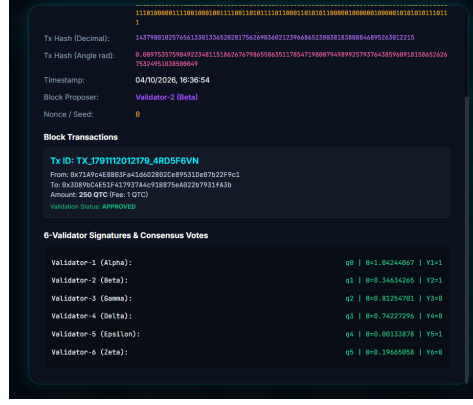


Fig. 6: Block Validator Information

The prototype successfully demonstrates an end to end transaction validation and block commitment cycle over the 6-node validator network after syncing up with the genesis block. As shown in the figure, when the submitted transaction has been validated with the blockchain protocol, the transaction's SHA-256 hash results into a hash prefix and decimal representation. In this case, we get 0.0097535759 rad as our computed transaction angle. Quantum validation is performed on the Delta node, where a random selection process designates the verifier for the current validation round. Subsequently, six measurement angles are generated such that their combined sum satisfies the required constraint $\sum_{i=1}^6 \Theta_i = \Pi$ radians. The six validator nodes produce the measurement vector as in Eq 15

$$\mathbf{Y} = [1, 1, 0, 0, 1, 0] \quad (15)$$

The measured XOR parity is calculated as

$$P_{\text{measured}} = 1 \oplus 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 1 \quad (16)$$

Since the measured parity in Eq16 matches the expected parity, $P_{\text{expected}} = 1$, the Θ -protocol consensus condition is satisfied, allowing the transaction to be accepted for block construction, as illustrated in Figures 5 and 6.

Block 1 is subsequently constructed by the Beta node and linked to the genesis block through its previous-block hash reference. Following synchronization across the six validator ledgers, each ledger incorporates the new block, increasing its height from 1 to 2, as illustrated in Figure 4. This was repeated for another 19 times and all the 20 independent simulations trials of the quantum validation achieved a 100% parity since it was in an ideal environment, and the average execution time was $3.25 \pm 0.05S$ per trial.

4 Discussion

The current work is simulator based and does not demonstrate practical security or real-world quantum consensus. While demonstrating successful performance by GHZ

based Validation using the above mentioned protocol we found out some constraints which are very crucial from the point of view of actual implementations. First, we see that the Verification condition (collective sum) requires the measurements angles should satisfy certain multiple of π i.e., $\theta_j = \frac{2\pi k_j}{N}$. An ideal simulator could ensure this constraint. However, real world Quantum Hardware suffers from finite precision control gates and cumulative angle errors leading to possible deviation in the effective sum θ_j thus reducing the reliability of desired parity output. Second, even though the proposed framework provides us with an encoding mechanism for representing the blockchain record as a entangle quantum state, the proposed Θ -protocol only verify the GHZ correlation of the candidate state, instead of verifying the presence of the intended block information in the state. As per the paper’s discussion they have described about how their classical block records were encoded into temporal Bell and GHZ States, but the part where you test if your supplied candidate State has the GHZ Correlations required is the Verification Stage here. So basically they showed off their Quantum Validation Mechanism. However, to build up a full-fledged Block Authentication Mechanism, we need to explicitly bind the classical block information we want to authenticate with the Candidate State to be Verified. This becomes really important because someone could simply go onto the Classical Blockchain Network, and tamper the block contents while keeping the underlying Quantum Consensus untouched. And since the Quantum Consensus doesn’t care about what happened on the classical side of things, it just happily accepts the tampered block. Now the recent paper regarding Time Entanglements in 2026 [10] manages to overcome this binding challenge, however it runs into serious challenges as the Number Of Blocks Increase, hence working out ways to Combine both methodologies may form an open research problem in our near future scope of Research.

Author contributions

Kasilingam, Alfred Sam D, and Brijith Manikandan contributed equally to the conception, implementation, experimentation, and manuscript drafting. V.Karthick supervised the project, provided technical guidance, and reviewed the final manuscript.

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Data availability

The data that was analyzed and implementation code for user interface and code implemented using QISKIT are available for academic use.

Declarations

Clinical trial number

Not applicable.

Ethics approval and consent to participate

This article does not contain any studies with human participants or animal performed by any of the authors. Hence, ethics approval was not required.

Consent for publication

Not applicable

Competing Interests

The authors declare no competing interests.

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